ORLANDO HEALTH®
1414 Kuhl Ave.
Orlando, FL 32806

DEPARTMENT GUIDELINE

Guideline Title: **Pediatric Diabetic Ketoacidosis (DKA)**	Guideline Number: **GUID-3203-PED008**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To recognize the presence and the precipitating factors of Diabetic Ketoacidosis (DKA), treat metabolic and hormonal abnormalities, and manage systemic complications as they present.

PATHOPHYSIOLOGY:

Diabetic ketoacidosis is characterized by hyperglycemia and ketonemia, resulting in an anion gap metabolic acidosis. Ketonemia occurs because of increased lipolysis and ketogenesis used by the cell to produce energy when glucose is not available. These patients are significantly dehydrated secondary to osmotic diuresis from increased glucose in their blood and have frequently have significant electrolyte imbalances. Rapid correction of hyperglycemia can produce CNS dysfunction, brain edema, and brain herniation.

Hyperglycemic Hyperosmolar Syndrome (HHS) is an emergency that is distinguished from DKA by marked hyperglycemia, absent to mild ketosis with minimal acidosis, and elevation in serum osmolality.

DEFINITIONS:

II. **DEPARTMENT GUIDELINE:**

A. Assess airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*.
B. Apply cardiac monitor, monitor pulse, blood pressure, and oximetry. Assess vital signs.
C. Apply oxygen for hypoxia. Monitor ETCO2 if intubated. Patients with acidosis must be adequately ventilated to aid in the correction of acidosis.
D. Obtain IV access for the patient's size.
E. Assess urine output.
F. Assess for Severe DKA:
   i. Lab findings: Hyperglycemia (blood glucose > 300-1,000 mg/dL.) Metabolic acidosis (pH < 7.1, HCO3 < 10 mEq/L). Elevated BUN and creatinine, normal to hyperkalemia and hyponatremia.
   ii. Physical exam: Profound Kussmaul's respirations, poor perfusion, severe dehydration with cool extremities, lethargy/obstruction, & poor urine output.
G. Hydration
   i. If signs of shock, follow *GUID3203-PED009 – Hypotension*.
   ii. If not in shock with signs of volume depletion, give 0.9% NS 20 ml/kg IV over 1 hour.
   iii. If no signs of volume depletion, start 0.9% NS at 1.5-2 times the maintenance rate.

1. Normal maintenance fluid rate:
    a. 4 mL/kg/hr IV for the first 10 kg.
    b. 2 mL/kg/hr IV for the second 10 kg
    c. 1 mL/kg/hr IV for each kg of body weight above 20 kg

H. Discuss with physician for orders for a **regular insulin infusion at 0.05-0.1 units/kg/hour** (no bolus).

I. Monitor blood glucose level every 30 minutes, adjust therapy to decrease serum glucose by no more than 100 mg/dL per hour.
    i. If serum glucose decreases by more than 100 mg/dL per hour, add 5% dextrose to IV fluids.
    ii. If serum glucose falls below 300 mg/dL, add 5% dextrose to the IV fluids.

J. Serum potassium may require supplementation via physician order **KCL IV infusion at 10-20 mEq/h**.

K. If pH is less than 7.0, contact Medical Control or speak to admitting physician to consider sodium bicarbonate.

L. Monitor for signs of cerebral edema (headache, decreased mental status)
    i. Elevate head of bed 30 degrees
    ii. Consider deceasing fluids and insulin infusion rates.

III. **DOCUMENTATION:** EMS Charts

IV. **REFERENCES:**

A. Glaser, Barnett, McCaslin, et al. (2001). Risk factors for cerebral edema in children with diabetic ketoacidosis. The Pediatric Emergency Medicine Collaborative Research Committee of the American Academy of Pediatrics. N Engl J Med; 344:264.

B. Wolfsdorf, Glaser, Agus, Fritsch, Hanas, Rewers, Sperling, & Codner. (2018). ISPAD Clinical Practice Consensus Guidelines 2018: Diabetic ketoacidosis and the hyperglycemic hyperosmolar state. *Pediatr Diabetes*, 19 Suppl 27:155.